

CATALINA, CA, USA

WMO: 722920

Lat: 33.405N

Lon: 118.416W

Elev: 488

StdP: 95.59

Time Zone: -8.00 (NAP)

Period: 00-19

WBAN: 23191

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.0	6.9	-15.5	1.0	18.7	-13.0	1.3	18.2	12.6	8.8	11.5	8.6	4.8	280	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	7.4	32.5	17.2	30.9	16.9	29.2	16.1	20.2	27.7	19.4	26.6	18.6	25.6	2.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	13.4	22.7	16.8	12.7	21.6	16.1	12.1	20.6	60.0	27.7	57.0	26.5	54.3	25.4	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.4	7.3	4.5	36.2	0.9	2.1	3.8	37.7	3.3	38.9	2.8	40.1	2.1	41.6		
(5)			0.9	21.6	0.9	0.9	0.3	22.3	-0.2	22.8	-0.7	23.3	-1.4	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	13.5	12.6	13.5	13.9	15.2	17.4	20.3	21.2	21.2	18.6	15.9	12.9	(6)
(7)		DBStd	5.22	4.10	4.21	4.05	4.02	4.04	4.01	4.00	4.22	4.76	4.76	4.34	3.52	(7)
(8)		HDD10.0	38	9	13	6	2	0	0	0	0	0	0	1	7	(8)
(9)		HDD18.3	1213	159	168	161	149	119	65	23	15	21	57	102	173	(9)
(10)		CDD10.0	2368	118	87	116	118	163	223	319	347	336	266	178	98	(10)
(11)		CDD18.3	499	9	9	13	14	23	38	84	103	107	65	28	6	(11)
(12)		CDH23.3	3777	10	17	47	90	171	322	636	880	967	495	137	5	(12)
(13)	CDH26.7	1159	0	0	5	20	48	114	184	273	348	143	24	0	(13)	
(14)	Wind	WSAvg	3.1	3.5	3.7	3.7	3.7	3.2	2.5	2.4	2.5	2.7	3.0	3.2	3.5	(14)
(15)	Precipitation	PrecAvg	295	56	62	40	19	7	2	2	1	6	15	29	52	(15)
(16)		PrecMax	696	201	190	95	45	27	7	15	3	26	89	51	179	(16)
(17)		PrecMin	129	4	0	12	4	0	0	0	1	4	0	5	0	(17)
(18)		PrecStd	154	63	54	28	14	8	2	4	1	5	25	14	53	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	25.5	27.4	29.2	31.2	33.8	33.8	33.8	35.1	32.8	29.4	23.9	(19)
(20)			MCWB	10.8	11.7	13.5	13.5	15.8	16.7	19.4	17.1	18.0	15.9	13.6	10.6	(20)
(21)		2%	DB	22.9	23.3	24.3	26.0	27.6	29.8	30.5	31.3	32.3	30.1	26.8	22.1	(21)
(22)			MCWB	10.4	10.7	12.0	12.8	14.4	16.3	18.2	17.2	17.3	14.9	12.6	9.9	(22)
(23)		5%	DB	21.4	21.6	22.4	23.1	24.8	27.0	28.5	29.5	30.5	28.0	24.8	20.2	(23)
(24)			MCWB	9.7	10.1	11.3	11.5	13.5	15.9	17.1	16.7	16.9	14.0	11.7	9.1	(24)
(25)		10%	DB	19.9	19.8	20.2	20.6	22.2	24.3	26.9	27.9	28.4	26.2	22.9	18.6	(25)
(26)			MCWB	9.2	9.3	10.4	11.1	12.8	15.1	16.8	16.1	15.8	13.1	10.9	8.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	14.7	15.4	16.4	17.7	19.7	21.2	21.2	21.0	18.7	15.8	14.6	(27)
(28)			MCDWB	18.1	17.7	21.8	24.1	26.9	29.4	29.9	28.6	29.8	25.3	20.1	16.5	(28)
(29)		2%	WB	13.2	13.4	14.1	14.5	16.2	18.0	20.1	20.0	19.7	17.5	14.9	13.6	(29)
(30)			MCDWB	16.0	17.0	19.8	21.3	23.8	26.4	27.4	27.3	27.1	23.7	19.3	15.8	(30)
(31)		5%	WB	12.2	12.2	13.1	13.4	15.2	17.1	19.2	19.0	18.7	16.5	14.0	12.6	(31)
(32)			MCDWB	15.7	16.7	18.4	19.4	21.4	24.0	25.9	26.1	26.6	22.4	18.3	15.2	(32)
(33)		10%	WB	11.3	11.3	12.2	12.5	14.2	16.3	18.3	18.2	17.7	15.6	13.2	11.7	(33)
(34)			MCDWB	15.2	15.6	17.0	17.3	19.0	22.3	24.5	24.8	24.8	20.8	17.9	14.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.3	5.4	6.3	6.8	6.7	7.5	8.1	7.9	7.4	6.5	5.9	5.1	(35)
(36)			MCDBR	6.6	7.6	8.5	10.0	9.7	10.2	9.6	8.4	8.9	8.2	7.4	6.9	(36)
(37)			MCWBR	4.1	4.8	5.3	5.6	5.8	6.2	6.5	6.6	5.8	5.0	4.2	4.2	(37)
(38)		5% WB	MCDWB	5.8	6.6	7.7	8.7	8.8	9.5	9.2	8.5	8.4	7.7	6.8	5.6	(38)
(39)			MCWBR	4.8	5.1	5.4	5.5	5.4	5.8	6.3	6.5	6.2	5.6	4.8	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.291	0.304	0.320	0.323	0.337	0.339	0.352	0.341	0.326	0.324	0.292	0.292	(40)
(41)		taud		2.638	2.618	2.548	2.538	2.523	2.512	2.540	2.555	2.597	2.587	2.659	2.658	(41)
(42)		Ebn at Noon		939	957	960	964	947	940	925	935	941	920	930	919	(42)
(43)		Edh at Noon		74	84	97	103	106	106	103	100	92	86	72	70	(43)
(44)	All-Sky Solar Radiation	RadAvg		3.18	4.06	5.35	6.53	6.77	6.69	6.92	6.67	5.66	4.35	3.46	2.86	(44)
(45)		RadStd		0.22	0.35	0.31	0.36	0.58	0.51	0.49	0.33	0.36	0.40	0.27	0.22	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.03	N/A	N/A	N/A	+0.78	+0.77	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	+0.64	N/A	-1.62	+0.78	N/A	N/A	N/A	-118	+230	+109

Nomenclature: See separate page